

DIBYAJYOTI MOHANTY

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PROFILE

AI/ML engineer with hands-on experience building production-grade NLP, speech, and generative AI systems, including Retrieval-Augmented Generation (RAG), agentic workflows, and real-time speech pipelines. Skilled in computer vision tasks such as object detection, image segmentation, and facial recognition. Consistently delivers measurable outcomes: 80% QA workload reduction, 65% HR ticket deflection, 60% faster provisioning. Experienced in leading cross-functional teams and translating complex research into reliable, high-performance production systems.

SKILLS

Languages & Libraries: Python (5+ years) | Pandas | NumPy | Matplotlib | Seaborn | Scikit-learn | TensorFlow | PyTorch | Keras | OpenCV | Flask | Streamlit | torchvision | FastAPI

Databases: SQL | MySQL | MongoDB | Neo4j | GraphDB | Vector Databases (Qdrant, ChromaDB, FAISS)

AI / ML Frameworks: PyTorch | TensorFlow | Keras | Scikit-learn | Hugging Face Transformers | LangChain | LangGraph | BERT | Sentence Transformers

Machine Learning: Supervised Learning | Unsupervised Learning | Reinforcement Learning

Deep Learning: CNN | R-CNN | YOLO | Deep SORT | Image Classification | Object Detection | Semantic Segmentation | RNN | LSTM | Bi-LSTM | Transformers | LLMs

Generative AI: RAG | Agentic RAG | Multimodal RAG | Knowledge Graph RAG | Conversational Memory | Prompt Engineering | AI Agents | Fine-tuning | LLM Evaluation | LLM Deployment | AI Deployment | MLOps

Speech & Audio AI: ASR | TTS | Voice Cloning | Speaker Diarization | Speech-to-Text | Text-to-Speech

Domains: Computer Vision | NLP | LLMs | Generative AI | RAG | Time Series Forecasting | Anomaly Detection

GPU & AI Infrastructure: CUDA | cuDNN | NVIDIA Drivers | GPU Optimization | AI Inference | Model Deployment | Dependency Management | Multi-GPU Scheduling

Version Control & DevOps: Git | GitHub | Bitbucket | Docker | GitHub Actions | CI/CD | Linux

Cloud & MLOps: AWS (SageMaker, ECS, API Gateway) | GCP | Azure ML | MLflow | Kubeflow | Airflow

Tools: VS Code | Cursor | Jupyter Notebook | Postman | Conda | Git | NVIDIA Nsight

EDUCATION

B.Sc. Information Science and Telecommunication | Ravenshaw University (2022)

EXPERIENCE

SDE – AI Engineer, Movius (*Contract via Canarys: Feb 2025 – Jan 2026 | Full-time: Feb 2026 – Present*) Feb 2025 – Present

- Built ASR-OTT, a real-time communication application for people with speech and hearing impairments: callers receive live transcription of the other party's speech, can send text messages that are converted to speech for the other party to hear, and receive an AI-generated summary of the conversation at call end. Core stack: Faster-Whisper (ASR), StyleTTS2 (TTS), and Mistral-70B (summarization).
- Built extensive audio pre-processing for streaming packets sent to the ASR model and post-processing to produce accurate, well-formatted transcriptions; modularized the ASR pipeline to allow easy benchmarking and swapping between different speech-to-text models.
- Implemented voice cloning in the TTS pipeline to produce natural, human-like speech while keeping latency low.
- Designed an agentic workflow with dynamic, per-call prompting to generate accurate conversation summaries.
- Optimized GPU utilization across all AI services using a round-robin scheduling strategy to maximize throughput and resource efficiency.
- Built a context-aware reply system using Retrieval-Augmented Generation with NVIDIA Nemotron: stored past call transcriptions in a vector database, searched archived conversation memory for relevant past exchanges, and scored memory relevance with a composite formula (importance × recency × freshness + relevance boost); retrieval used cosine similarity with reranking to surface 3 candidate responses per query, letting the user pick the best fit.
- Managed and optimized the GPU AI infrastructure by configuring and maintaining compatibility across CUDA, cuDNN, NVIDIA drivers, PyTorch, TensorFlow, and NVIDIA libraries, ensuring stable deployment, maximum GPU utilization, and seamless execution of AI inference services.
- Worked with Bitbucket, Docker, CI/CD pipelines, and Git for source control and deployment.

Generative AI Engineer, ATG

Jan 2024 – Sep 2024

- Led the development of two end-to-end AI solutions as the Lead AI Engineer, leveraging Python, Machine Learning, LLMs, NLP, Docker, MLOps, and Generative AI technologies.
- Designed and implemented Retrieval-Augmented Generation (RAG) pipelines to improve contextual understanding, response accuracy, and knowledge retrieval for AI applications.

- Developed an AI-powered Sales Call Analysis platform that processed customer-sales conversation audio, performed speaker diarization (**Deepgram**) to identify the salesperson's speech, transcribed conversations, and utilized LLMs (**llama-3**) to evaluate sales performance by assessing response quality, customer query coverage, communication effectiveness, and identifying missed opportunities and areas for improvement.

AI/ML Consultant, **Freelance**

Feb 2023 – Dec 2023

- Delivered end-to-end AI and Generative AI solutions for multiple clients, developing production-ready applications using Python, LLMs, NLP, Computer Vision, RAG, and MLOps.
- Developed AI-powered conversational systems capable of real-time audio analysis by integrating Gemini, Telegram Bot API, ASR, TTS, and NLP pipelines for intelligent speech understanding and voice-enabled interactions.
- Built AI applications leveraging OpenAI DALL·E, Flask, MongoDB, and U-Net-based medical image segmentation models for image generation and healthcare computer vision use cases.
- Fine-tuned Qwen 3.5 Vision-Language Models (VLMs) using QLoRA with the Unsloth framework, utilizing datasets from Hugging Face to optimize multimodal reasoning.
- Implemented Human-in-the-Loop (HITL) reinforcement learning workflows to iteratively improve VLM performance through human feedback, enhancing model accuracy, response quality, and alignment with task-specific requirements.

Programmer Analyst, **Cognizant**

Aug 2022 – Jan 2023

- Performed data analysis, data transformation, and data pre-processing using Python and MySQL to prepare high-quality datasets for machine learning and business analytics.
- Contributed to data engineering initiatives by developing and optimizing data pipelines, improving data quality, processing efficiency, and system performance.
- Collaborated with cross-functional teams to extract, validate, and analyze large datasets, delivering actionable insights to support data-driven decision-making.

Software Developer, **Stack System**

Feb 2020 – Aug 2021

- Developed and enhanced backend features for a Flask-based responsive collage web application, implementing scalable RESTful APIs and optimizing application performance using Python.
- Mentored graduate students in Python programming, software development best practices, and problem-solving techniques through hands-on training and code reviews.

PROJECTS

AI-Powered NPC System with Dynamic Personality & Memory Architecture (In Progress)

- Architected multi-layered AI agent system featuring persona core (traits, backstory, motives), perception layer (context awareness), decision engine (emotions, goals, memory), and LLM integration (Groq/Llama 3.3 70B) for real-time dialogue generation
- Implemented dual-memory system with short-term (recent 10 interactions) and long-term (importance-weighted) storage, enabling NPCs to recall past conversations, track relationship trust scores (-100 to +100), and adapt responses based on emotional state
- Developed emergent social dynamics including NPC-to-NPC conversations, gossip propagation networks, dynamic world events system, and action intent framework allowing autonomous agent behaviors (investigate, confront, flee, call guards)

Multi-Agent AI Assistant with Long-Term Memory

- Designed and developed a multi-agent conversational AI system using LangChain, LangGraph, Groq Llama-3, and Flask/React, supporting threaded conversations and dynamic context switching.
- Implemented agent orchestration, tool routing, and memory management to enable collaborative reasoning and context-aware dialogue.

Enterprise Conversation Intelligence & Business Analyst Evaluation:

- Developed an AI-powered conversation intelligence platform that analyzed client-business analyst interactions using ASR, speaker diarization, and RAG to evaluate response quality and knowledge accuracy.
- Built a retrieval pipeline using Google Embedding-Text-004, ChromaDB, and semantic chunking to ground LLM responses against enterprise documentation.
- Leveraged Deepgram Nova-2 for speaker diarization and transcription, followed by Groq Llama-3 for contextual evaluation, feedback generation, and performance scoring.